Firefly

Offline incident reporting for wildfire crews



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Radio congestion

> “excessive radio traffic on the initially assigned tactical frequency, leading to delayed communication”

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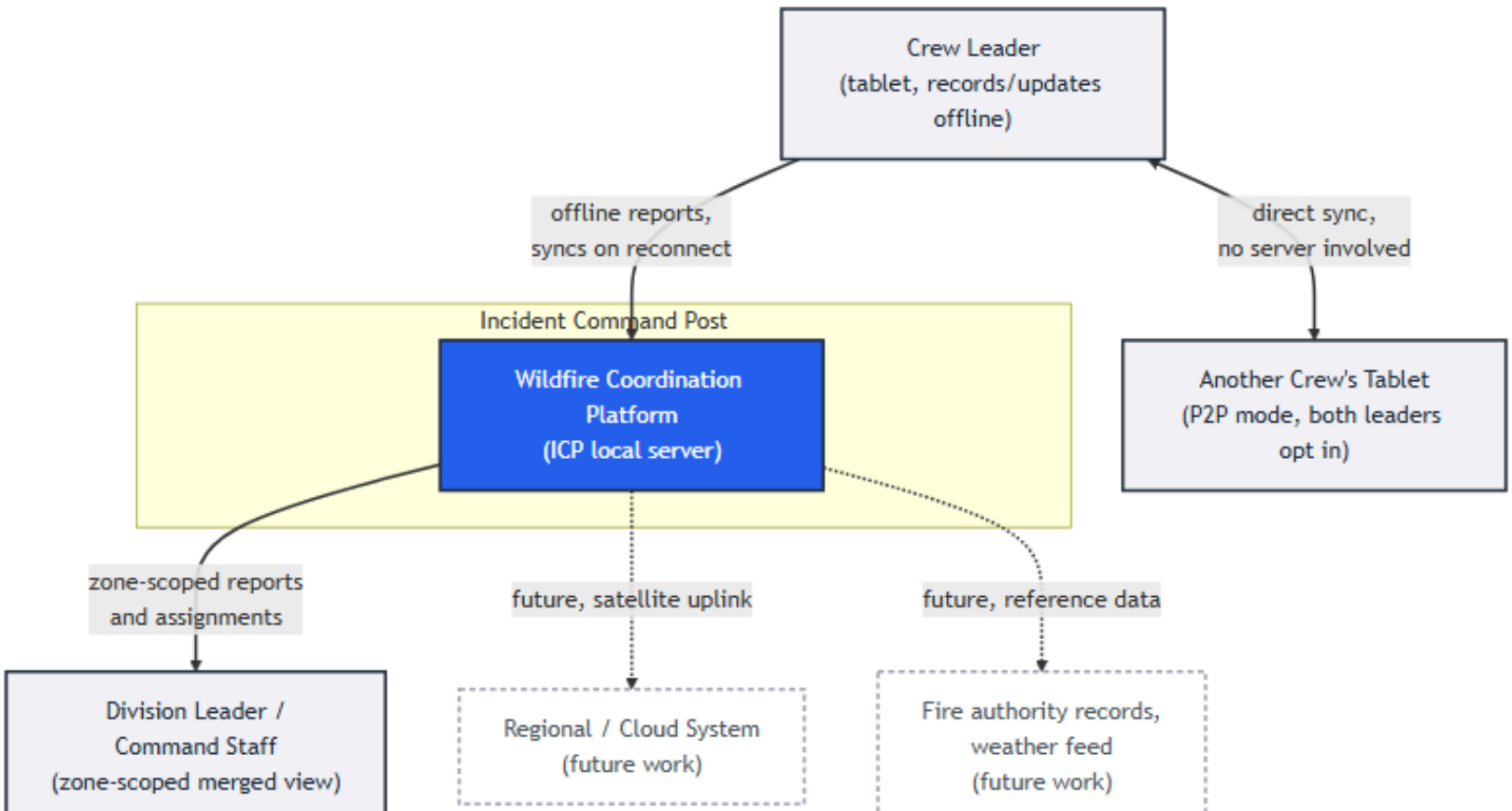
people killed at Überlingen, 2002.
One controller, two workstations.


Limited connectivity in the field

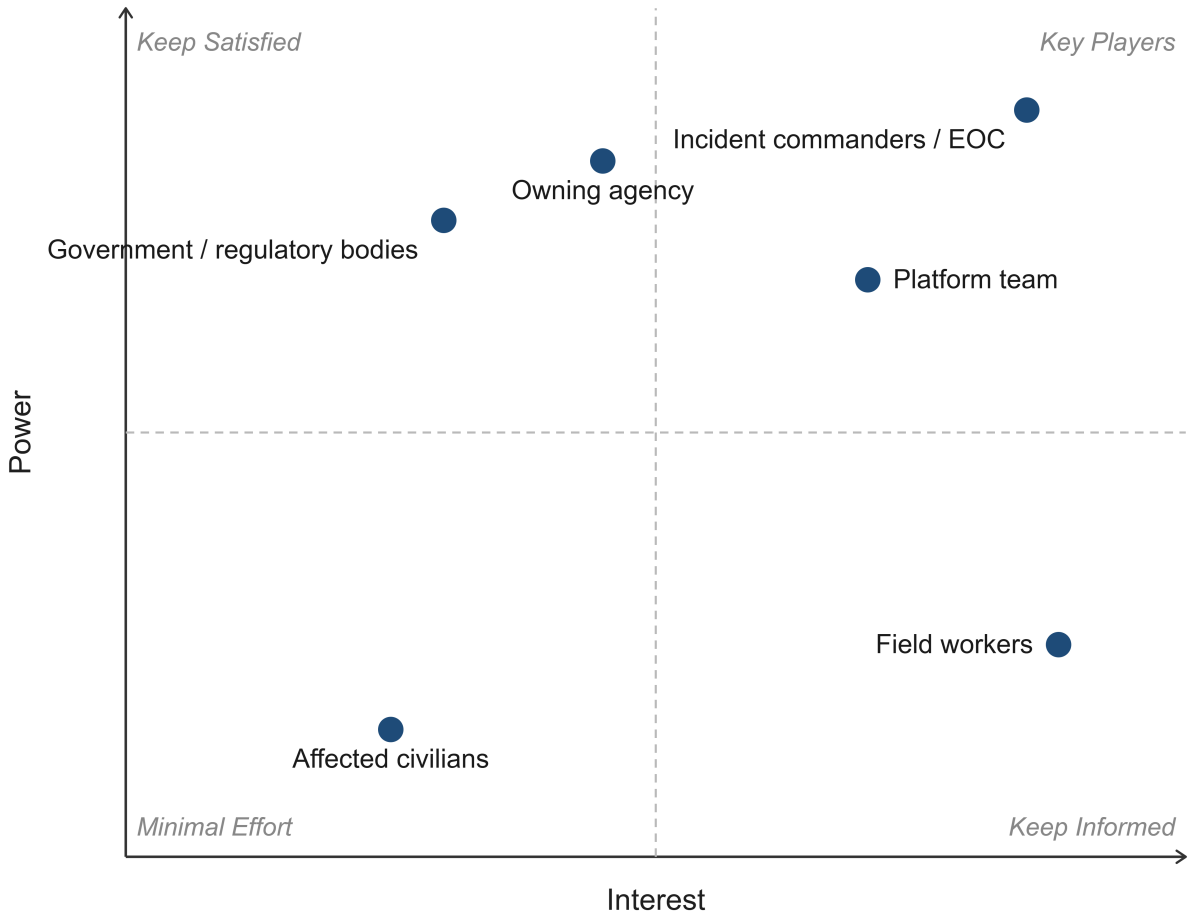
Existing tools

ArcGIS Field Maps	Works offline. Last write wins; conflicts go to a database admin.
ATAK + goTenna or Meshtastic	Device to device. No merging of concurrent edits.

System context



Stakeholders



Scenarios

Connected

Alice marks a blocked road. Command sees it.

Offline

Alice logs a spot fire below the ridge.

Crews meet

Alice and Bob switch on P2P mode.

Back in coverage

Bob’s tablet brings both crews’ reports to command.

Conflicts

Same access road, edited offline by two crews

Alice	Blocked
Bob	Open

Both reports are kept, with author and time, until the crew leaders resolve the conflict.



Architecturally significant requirements

R1 Store, update and sync in P2P mode while offline

R2 Sync with the server whenever connected

R3 Any two peers, server included, end with the same record

R4 No report is lost silently

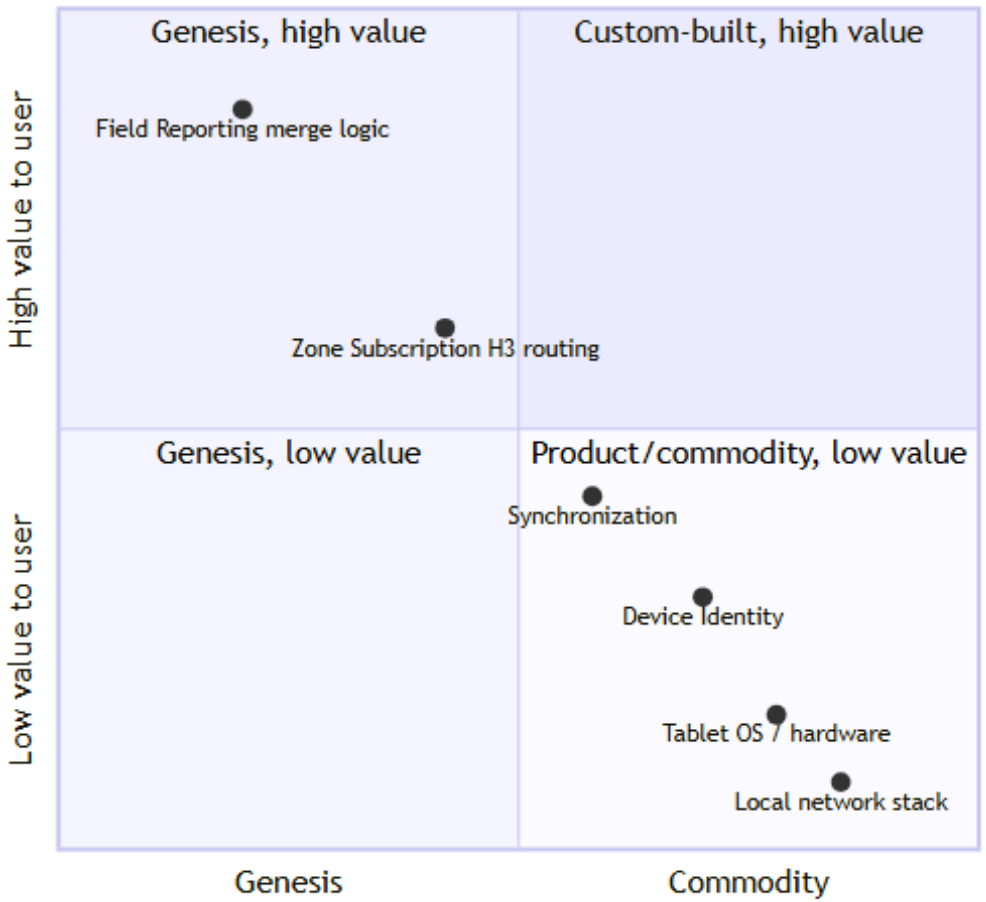
R5 P2P needs both crew leaders to opt in

R6 Server sends only reports from a tablet’s zones

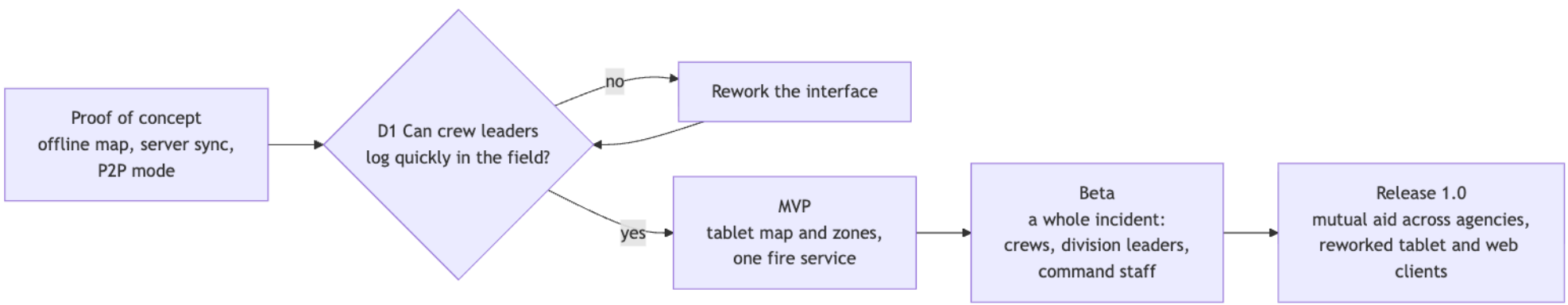
Quality attributes

Attribute	Target
Availability under partition	12 hours fully usable offline
Reliability	0 reports lost under injected failures
Scalability	Per-tablet load flat from 1 crew to dozens (beta)
Compatibility	One version apart still syncs

Wardley map


Genesis, high value
Custom-built, high value
Genesis, low value
Product/commodity, low value
High value to user
Low value to user
Field Reporting merge logic
Zone Subscription H3 routing
Synchronization
Device Identity
Tablet OS / hardware
Local network stack
Genesis
Commodity

Roadmap


Proof of concept
offline map, server sync,
P2P mode
D1 Can crew leaders
log quickly in the field?
no
Rework the interface
yes
MVP
tablet map and zones,
one fire service
Beta
a whole incident:
crews, division leaders,
command staff
Release 1.0
mutual aid across agencies,
reworked tablet and web
clients

Pricing model

One licence per agency, any number of tablets

Free until release 1.0

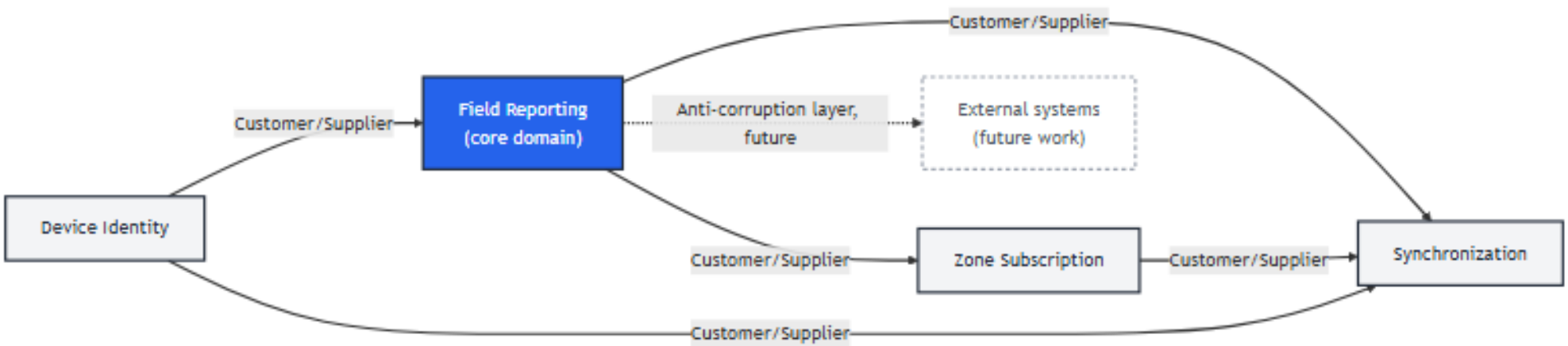
Yearly support fee, a percentage of the licence

Deployment and integration billed separately

Ethical and legal considerations

Automation bias	Always show that a merge was automatic
False reports	Authenticate users and devices
Personal data	Limit retention and access to merge history (GDPR)
Older hardware	Runs on older tablets

Bounded contexts



Merge logic

Merge logic 1: domain events

Event	Operation
Report created	`CREATE`
Report field updated	`SET_FIELD`
Field marked conflicted	derived
Conflict resolved by a crew leader	`RESOLVE_CONFLICT`
Two tablets synced in P2P mode	sync
Report routed to subscribed tablets	delivery

Merge logic 2: merge tiers

Tier A, automatic

Severity maximum
Report type union
Incident name fixed tiebreak

Tier B, crew leader

Location
Description
Incident status

Dotted version vectors detect concurrent writes.

Merge logic 3: synchronization

Tablet to server	On reconnect, the tablet uploads everything logged offline, and the server merges it
Server to tablet, on reconnect	The tablet fetches the reports it missed for its zones
Server to tablet, while online	The server pushes new reports for the tablet's zones as they arrive

Current progress

Report	Problem analysis written, architecture in first draft
Proof of concept	Shared merge core (severity so far), server skeleton, device stub
Next	Server sync and P2P between two tablets